Lecture 17: Summing Infinite Series II

Telescoping Series

A telescoping series is a special type of series for which most terms canceled in the nth partial sums.

To determine whether a telescoping series converges, we again write out the nth partial sums.

If a telescoping series converges, we are able to find the **exact sum**.

$-1 < r < 1$

① Geo series $\sum ar^n = \frac{\text{first term}}{1-r}$

② telescoping series.

1. Find Partial Fraction.
   (if a_n is a rational expression)
2. Write out the partial sum S_n
3. Find $\lim_{n\to\infty} S_n$

TFD: $\lim_{n\to\infty} a_n \neq 0 \rightarrow \sum a_n$ div.

Determine if the following series converge.
If converge, find the sum:

$\lim_{n\to\infty} \frac{1}{n(n+1)} = 0$

ex. 1. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ conv. to 1.

$$a_n = \frac{1}{n(n+1)} = \frac{A=1}{n} + \frac{B=-1}{n+1} = \frac{1}{n} - \frac{1}{n+1}$$

$$\begin{aligned} S_N = {} & 1 - \cancel{\tfrac{1}{2}} && (n=1) \\ & + \cancel{\tfrac{1}{2}} - \cancel{\tfrac{1}{3}} && (n=2) \\ & + \cancel{\tfrac{1}{3}} - \cancel{\tfrac{1}{4}} && (n=3) \\ & \vdots \\ & + \cancel{\tfrac{1}{N}} - \tfrac{1}{N+1} && (n=N) \end{aligned}$$

$$S_N = 1 - \frac{1}{N+1}$$

$$\text{Sum} = \lim_{N\to\infty} 1 - \frac{1}{N+1} = 1$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1}$$

conv.

If $\sum a_n$ and $\sum b_n$ div.

$\Rightarrow \sum(a_n+b_n) \neq \sum a_n + \sum b_n$

$\sum_{n=1}^{\infty} \frac{1}{n}$ div.

$\sum_{n=1}^{\infty} \frac{1}{n+1}$ div.

$$\lim_{n\to\infty} \ln\left(\frac{n}{n+1}\right) = \ln(1) = 0.$$

ex. 2. $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$. div.

$$a_n = \ln\left(\frac{n}{n+1}\right) = \ln(n) - \ln(n+1)$$

$$\begin{aligned}
S_N = \ln 1 - \cancel{\ln 2} \qquad & (n=1)\\
+ \cancel{\ln 2} - \cancel{\ln 3} \qquad & (n=2)\\
+ \cancel{\ln 3} - \cancel{\ln 4} \qquad & (n=3)\\
\vdots \qquad & \\
+ \cancel{\ln N} - \ln(N+1) \qquad & (n=N)
\end{aligned}$$

$$S_N = \ln 1 - \ln(N+1) = -\ln(N+1)$$

$$\text{sum} = \lim_{N\to\infty} -\ln(N+1) = -\infty$$

$$\ln(N+1) = \ln\frac{1}{N+1} \to \ln 0^+$$

div.: $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right) = \sum_{n=1}^{\infty} \ln n - \ln(n+1)$

$$\begin{cases} \displaystyle\sum_{n=1}^{\infty} \ln n : \text{div. (TFD)} \\ \displaystyle\sum_{n=1}^{\infty} \ln(n+1) : \text{div. (TFD)} \end{cases}$$

Theorem: If $\sum a_n$ and $\sum b_n$ are $\boxed{\text{both convergent}}$ series, then so are $\sum ca_n$, $\sum(a_n \pm b_n)$, and

$$\sum ca_n = c\sum a_n \quad (c \text{ any constant}),$$

$$\sum(a_n \pm b_n) = \sum a_n \pm \sum b_n$$

ex. 3. $\displaystyle\sum_{n=1}^{\infty}\left(\frac{3}{n(n+3)} + \frac{1}{2^n}\right)$ conv. to $\frac{11}{6} + 1 = \boxed{\frac{17}{6}}$

① $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n(n+3)}$: $a_n = \dfrac{3}{n(n+3)} = \dfrac{A=1}{n} + \dfrac{B=-1}{n+3} = \dfrac{1}{n} - \dfrac{1}{n+3}$

$$= \underbrace{\frac{1}{n} - \frac{1}{n+1}} + \underbrace{\frac{1}{n+1} - \frac{1}{n+2}} + \underbrace{\frac{1}{n+2} - \frac{1}{n+3}}$$

$$\begin{aligned}
S_N = {} & 1 - \cancel{\tfrac{1}{2}} + \tfrac{1}{2} - \cancel{\tfrac{1}{3}} + \tfrac{1}{3} - \cancel{\tfrac{1}{4}} && (n=1)\\
& + \cancel{\tfrac{1}{2}} - \cancel{\tfrac{1}{3}} + \cancel{\tfrac{1}{3}} - \cancel{\tfrac{1}{4}} + \cancel{\tfrac{1}{4}} - \cancel{\tfrac{1}{5}} && (n=2)\\
& + \cancel{\tfrac{1}{3}} - \cancel{\tfrac{1}{4}} + \cancel{\tfrac{1}{4}} - \cancel{\tfrac{1}{5}} + \cancel{\tfrac{1}{5}} - \cancel{\tfrac{1}{6}} && (n=3)\\
& \vdots \\
& + \cancel{\tfrac{1}{N}} - \tfrac{1}{N+1} + \cancel{\tfrac{1}{N+1}} - \tfrac{1}{N+2} + \cancel{\tfrac{1}{N+2}} - \tfrac{1}{N+3} && (n=N)
\end{aligned}$$

$$S_N = 1 + \frac{1}{2} + \frac{1}{3} - \underset{\downarrow\, 0}{\frac{1}{N+1}} - \underset{\downarrow\, 0}{\frac{1}{N+2}} - \underset{\downarrow\, 0}{\frac{1}{N+3}}, \quad \text{sum} = 1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$$

② $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$

NYTI:

1. show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$ converges and find the sum it converges to.

$$a_n = \frac{2}{n^2+4n+3} = \frac{2}{(n+1)(n+3)} = \frac{A=1}{n+1} + \frac{B=-1}{n+3}$$

$$= \underbrace{\frac{1}{n+1} - \frac{1}{n+2}} + \underbrace{\frac{1}{n+2} - \frac{1}{n+3}}$$

$$\begin{aligned} S_N = {} & \frac{1}{2} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} && (n=1) \\ & + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{4}} - \cancel{\frac{1}{5}} && (n=2) \\ & + \cancel{\frac{1}{4}} - \cancel{\frac{1}{5}} + \cancel{\frac{1}{5}} - \cancel{\frac{1}{6}} && (n=3) \\ & \vdots \\ & + \cancel{\frac{1}{N+1}} - \frac{1}{N+2} + \cancel{\frac{1}{N+2}} - \frac{1}{N+3} \end{aligned}$$

$$S_N = \frac{1}{2} + \frac{1}{3} - \frac{1}{N+2} - \frac{1}{N+3}$$

$$\text{Sum} = \lim_{N\to\infty} S_N = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

2. Use partial fractions to find the sum of the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^3-n}$. $\left(\frac{1}{4}\right)$

$$a_n = \frac{1}{n(n-1)(n+1)} = \frac{A=\frac{1}{2}}{n-1} + \frac{B=-1}{n} + \frac{C=\frac{1}{2}}{n+1}$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{n-1} - \frac{1}{n}} - \underbrace{\frac{1}{n} + \frac{1}{n+1}}\right]$$

$$\begin{aligned} S_N = \frac{1}{2}\Big[& 1 - \cancel{\frac{1}{2}} - \frac{1}{2} + \cancel{\frac{1}{3}} && (n=2) \\ & + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{4}} && (n=3) \\ & + \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{5}} && (n=4) \\ & \vdots \\ & + \cancel{\frac{1}{N-1}} - \frac{1}{N} - \cancel{\frac{1}{N}} + \frac{1}{N+1}\Big] \end{aligned}$$

$$S_N = \frac{1}{2}\left[1 - \frac{1}{2} - \frac{1}{N} + \frac{1}{N+1}\right]$$

$$\text{Sum} = \lim_{N\to\infty} S_N = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}.$$

3. Find the sum of the series $\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$.

$$a_n = \ln\left(1 - \frac{1}{n^2}\right) = \ln\left(\frac{n^2-1}{n^2}\right) = \ln\left[\frac{(n-1)(n+1)}{n^2}\right]$$

$$= \ln(n-1) + \ln(n+1) - \ln n^2$$

$$= \ln(n-1) - 2\ln n + \ln(n+1)$$

$$= \underbrace{\ln(n-1) - \ln n}\ \underbrace{- \ln n + \ln(n+1)}$$

$$S_N = \ln 1 - \cancel{\ln 2} - \ln 2 + \cancel{\ln 3} \quad (n=2)$$

$$+ \cancel{\ln 2} - \cancel{\ln 3} - \cancel{\ln 3} + \cancel{\ln 4} \quad (n=3)$$

$$+ \cancel{\ln 3} - \cancel{\ln 4} - \cancel{\ln 4} + \cancel{\ln 5} \quad (n=4)$$

$$\vdots$$

$$+ \cancel{\ln(N-1)} - \ln N - \cancel{\ln N} + \ln(N+1)$$

$$S_N = \cancel{\ln 1}^{\,0} - \ln 2 - \ln N + \ln(N+1)$$

$$= -\ln 2 + \ln\left(\frac{N+1}{N}\right)$$

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} -\ln 2 + \ln\left(\frac{N+1}{N}\right) = -\ln 2 + \ln 1 = -\ln 2$$

4. Find the values of x for which the following series converges. Find the sum of the series for these values of x.

$$\sum_{n=0}^{\infty} \frac{(x+3)^n}{2^n}$$: Geo. series with $r = \frac{x+3}{2}$

$(-5 < x < -1$, converges to $\frac{2}{-1-x})$

$-1 < \frac{x+3}{2} < 1$

$-2 < x+3 < 2$

$-5 < x < -1$

$$\text{sum} = \frac{1}{1-\frac{x+3}{2}} = \frac{1}{\frac{2-x-3}{2}}$$

$$= \frac{2}{-1-x}$$

5. Suppose both $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ diverge. Then $\sum_{n=1}^{\infty}(a_n - b_n)$ diverges. True or False?

$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$ conv. to 1.

$\sum_{n=1}^{\infty} \frac{1}{n}$ and $\sum_{n=1}^{\infty} \frac{1}{n+1}$ both div.

see page 2.

6. Sum of a divergent and a convergent series must diverge. True or False?

$\sum a_n$ div. $\rightarrow \sum a_n = \lim S_N$ DNE.

$\sum b_n$ conv. $\rightarrow \sum b_n = \lim T_N = T < \infty$

$\sum(a_n+b_n)$? $\lim_{N\to\infty} U_N = \lim(S_N + T_N)$: DNE $\rightarrow \sum(a_n+b_n)$ div.

7. Determine the convergence of the series and find the sum if it converges.

$$\sum_{n=2}^{\infty} \frac{-1+(-3)^n}{4^n} = \sum_{n=2}^{\infty} \frac{-1}{4^n} + \frac{(-3)^n}{4^n}$$ conv. to $-\frac{1}{12}+\frac{9}{28}$.

$\sum_{n=2}^{\infty} \frac{-1}{4^n}$: $r=\frac{1}{4}$, sum $= \dfrac{\frac{-1}{16}}{1-\frac{1}{4}} = -\frac{1}{16}\cdot\frac{4}{3} = \frac{-1}{12}$

$\sum_{n=2}^{\infty} \frac{(-3)^n}{4^n} = \sum_{n=2}^{\infty} \left(\frac{-3}{4}\right)^n$: $r=-\frac{3}{4}$, sum $= \dfrac{\frac{9}{16}}{1-(-\frac{3}{4})}$

$= \frac{9}{16}\cdot\frac{4}{7}$

$= \frac{9}{28}$